End of Program Presentation

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Data Analytics Engineer Co-op

Agenda

1. About Me & Projects Overview
2. Operationalized AI Task Force Request Intake and Tracking
3. DBT models for Ariba and SRM process in accordance with the finance policy
   - Goal
   - Process
   - Key Takeaways
4. Premium Freight App
5. Lessons Learned and Next Steps

Introduction

About Me

Master of Science in Software Engineering Systems student at Northeastern University, Boston.
Data Analytics Engineer (Fall Co-op) at Bose.
Passionate about building scalable data models and pipelines using dbt and Snowflake, with a growing specialization in data analytics and analytical engineering.

Projects

Designed DBT models for Ariba Finance Policy and SRM feed: structured, validated DBT pipeline to stage, enrich, and produce accurate approval datasets for non-Ariba and later for Ariba, while ensuring strict column order and alignment with stakeholder expectations.

Premium Freight App: Built and owned an end-to-end customer-facing app using Streamlit and Snowflake, enabling supply chain teams to identify the root cause for air freight shipping orders

Automate AI Task Force SharePoint Intake Process: Built a Microsoft Power Automate workflow that streamlined and scaled Gen AI request management

Intermediate View for In-the-home connected products: A new dbt intermediate view (INT_CONNECTED_PRODUCTS_ITH) was defined by mapping key product, passport, account, and feature-usage attributes from multiple source models into a consolidated table for analytics.

AI Taskforce Request Intake

Goal

Goal:
Streamline and scale the GenAI request management process by replacing the current manual, fragmented intake workflow with a Microsoft-based solution (Forms, Lists, SharePoint) that delivers real-time visibility, parallel reviews, and transparent decision tracking.

Ask:
Adopt and implement this Microsoft Power Automate solution to reduce delays, improve oversight, and enable scalable governance of GenAI initiatives.

Current intake and tracking process is manual, fragmented, and lacks transparency

→ Microsoft-based solution can streamline and scale GenAI request management

Situation

- AI Task Force relies on a manual request intake process (copy/paste into board).
- Reviews (legal, security, solutioning) are tracked sequentially, not in parallel.
- Decisions and categories (e.g., coding assistants) are difficult to access.
- Updates primarily occur during AI TF meetings → information quickly becomes stale.

Complication

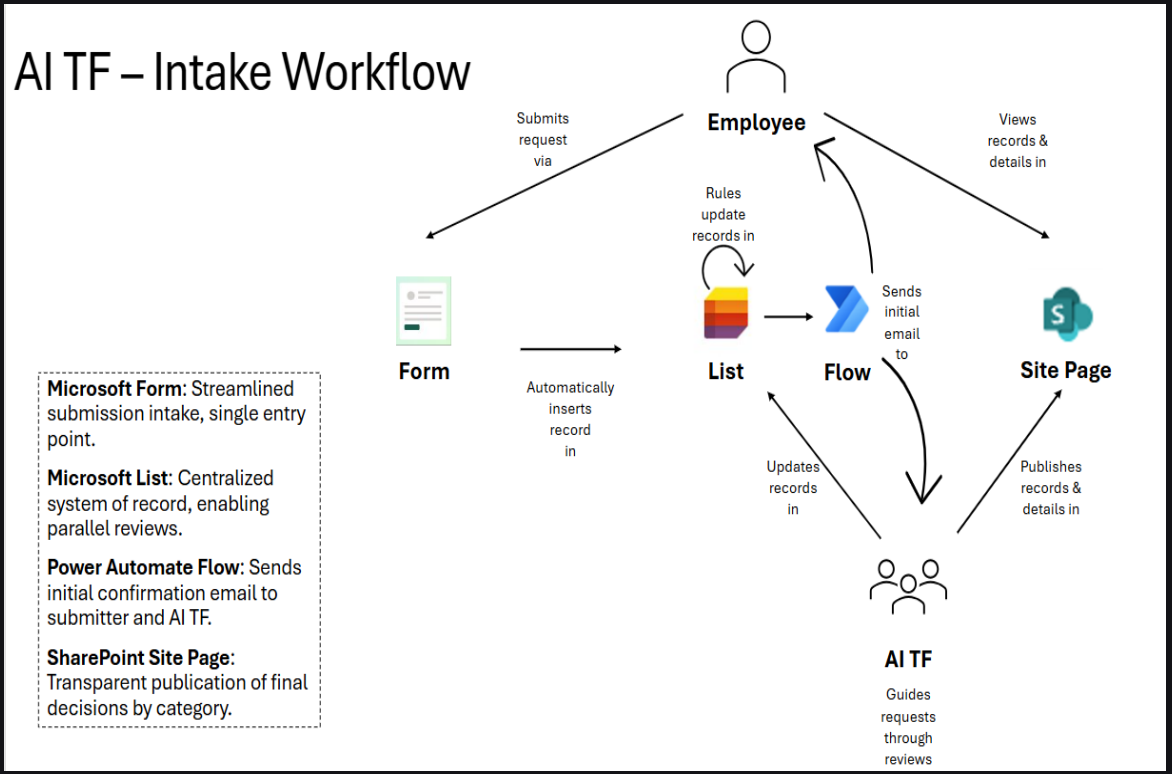
- Manual process creates delays, rework, and risk of missed requests.
- Lack of transparency reduces stakeholder confidence in decisions.
- Inability to easily view requests by category limits strategic oversight.
- Data is not real-time → executive decision-making relies on outdated info.

Resolution

- Microsoft Form: Streamlined submission intake, single entry point.
- Microsoft List: Centralized system of record, enabling parallel reviews.
- SharePoint Site Page: Transparent publication of final decisions by category.
- Outcome: Faster reviews, improved visibility, and scalable governance of GenAI initiatives.

AI Intake Workflow

- The workflow begins with an employee submitting a request through a **Microsoft Form**, which automatically creates a record in a **Microsoft List**, serving as the centralized system of record.
- A **Power Automate Flow** sends confirmation emails and updates list records as the AI Task Force completes reviews, enabling parallel tracking and real-time status updates.
- Final decisions and details are published to a **SharePoint Site Page**, giving employees transparent visibility into request outcomes and category-level insights.

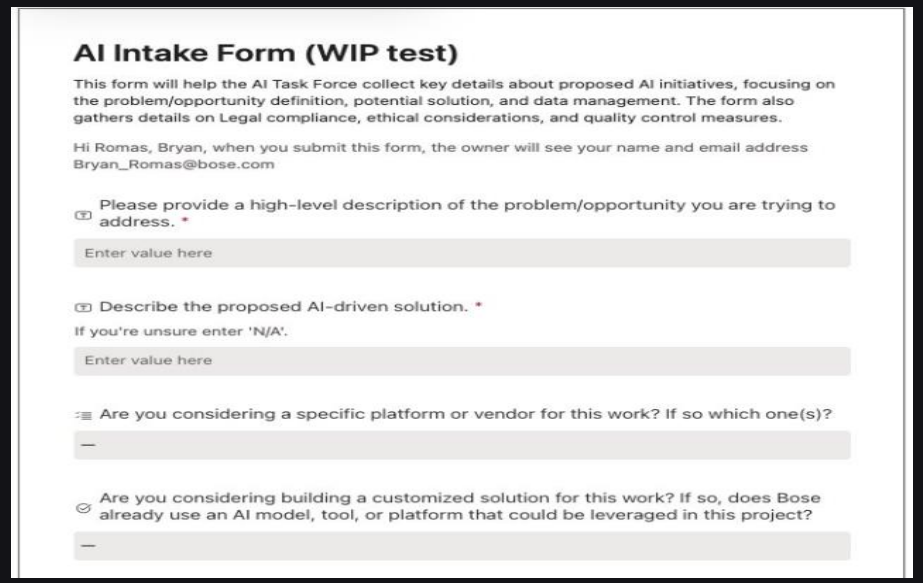


Solution Implemented

1. **Form:** Streamlines submission of GenAI requests and auto-creates records in Microsoft List (no manual copying). Serves as single-entry point for all records

2. **List:** Captures all GenAI requests captured from the AI Intake form and tracks review progress across Legal, Security and InfoSec. Serves as single source of truth for AI TF.

3. **Site Page:** Publishes finalized submissions and decisions for transparency. It acts as official reference hub for Gen AI adoption decisions and provides narratives & context for each category.



AI Intake Form (WIP test)

This form will help the AI Task Force collect key details about proposed AI initiatives, focusing on the problem/opportunity definition, potential solution, and data management. The form also gathers details on Legal compliance, ethical considerations, and quality control measures.

Hi Romas, Bryan, when you submit this form, the owner will see your name and email address Bryan_Romas@bose.com

Please provide a high-level description of the problem/opportunity you are trying to address. *

Enter value here

Describe the proposed AI-driven solution. *

If you're unsure enter 'N/A'.

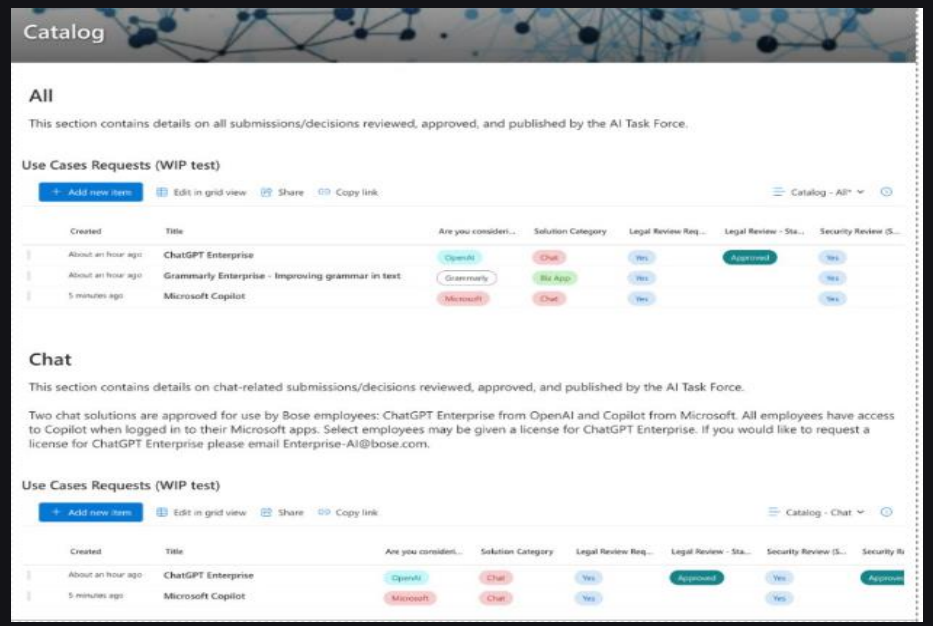
Enter value here

Are you considering a specific platform or vendor for this work? If so which one(s)?

—

Are you considering building a customized solution for this work? If so, does Bose already use an AI model, tool, or platform that could be leveraged in this project?

—



Catalog

All

This section contains details on all submissions/decisions reviewed, approved, and published by the AI Task Force.

Use Cases Requests (WIP test)

Add new item | Edit in grid view | Share | Copy link | Catalog - All

Created	Title	Are you consideri...	Solution Category	Legal Review Req...	Legal Review - Sta...	Security Review (S...
About an hour ago	ChatGPT Enterprise	OpenAI	Chat	Yes	Approved	Yes
About an hour ago	Grammarly Enterprise - Improving grammar in text	Grammarly	Biz App	Yes		Yes
5 minutes ago	Microsoft Copilot	Microsoft	Chat	Yes		Yes

Chat

This section contains details on chat-related submissions/decisions reviewed, approved, and published by the AI Task Force.

Two chat solutions are approved for use by Bose employees: ChatGPT Enterprise from OpenAI and Copilot from Microsoft. All employees have access to Copilot when logged in to their Microsoft apps. Select employees may be given a license for ChatGPT Enterprise. If you would like to request a license for ChatGPT Enterprise please email Enterprise-AI@bose.com.

Use Cases Requests (WIP test)

Add new item | Edit in grid view | Share | Copy link | Catalog - Chat

Created	Title	Are you consideri...	Solution Category	Legal Review Req...	Legal Review - Sta...	Security Review (S...	Security Re
About an hour ago	ChatGPT Enterprise	OpenAI	Chat	Yes	Approved	Yes	Approve
5 minutes ago	Microsoft Copilot	Microsoft	Chat	Yes		Yes	

Dbt models for Ariba and SRM process in accordance with the finance policy

Goal

Ariba & SRM -

The goal of this project was to replace a legacy process built in SQL server with a new process fully automated in Snowflake. This automation effort will provide us the ability to remove this legacy SQL Server saving cost for Bose while also optimizing the process to be more current with the latest technology stack & COE processes.

The newly designed process leverages Snowflake, supported by cloud-native automation and orchestration components that align with COE (Center of Excellence) best practices. Key aspects include:

1. Automated ingestion and transformation of Ariba & SRM data
2. Snowflake-based pipelines replacing manual SQL Server jobs
3. Centralized and version-controlled FP-1 file processing workflow
4. End-to-end validation and file delivery to S3
5. Integration with downstream systems for new/updated/deleted row processing

Process

This process describes how a requested change to an **FP-1 table** is validated, applied to a file, and then pushed to an S3 location so downstream systems can update new/modified/deleted rows.

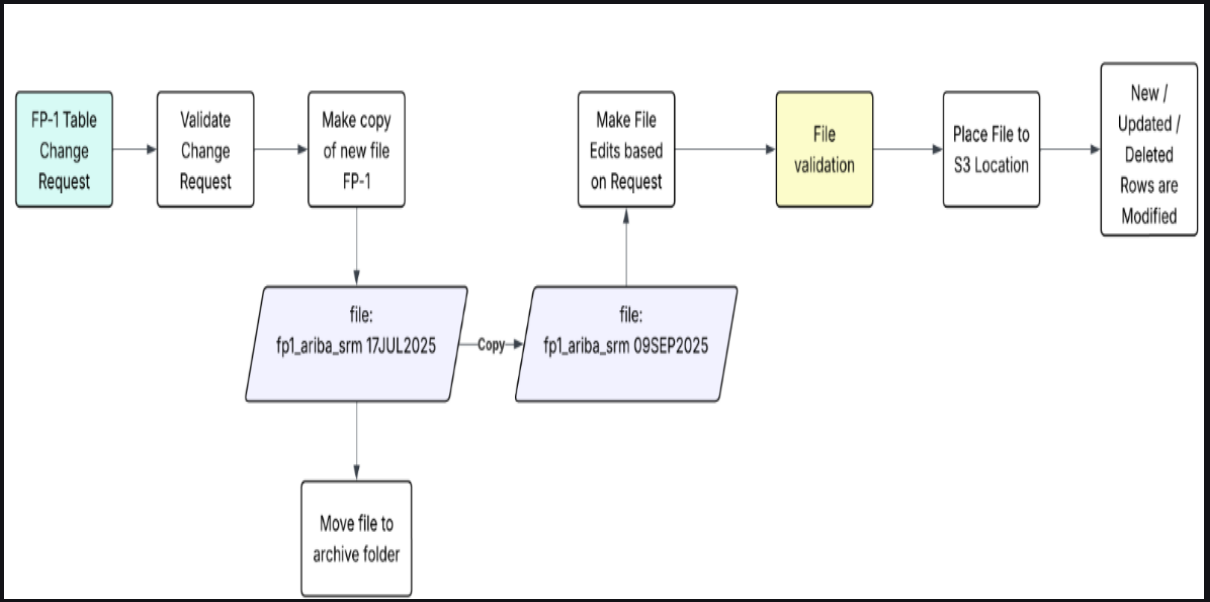
1. FP-1 Table Change Request
Stakeholders submit a request asking for changes to the FP-1 table.

2. Validate Change Request
The request is reviewed to ensure it is complete
If validation fails, the process stops here.

3. Make Copy of New FP-1 File
A fresh copy of the master FP-1 file is generated.

4. Copy Previous FP-1 File
The previously processed FP-1 version is copied to create the next version.

5. File is validated and final updated fp1 file is placed in s3 bucket.



Data Models Process

•Built **stg_ariba_srm** as a staging model on AID_ADP_PSA.FP1_ARIBA_SRM.RAW, pulling all fields and implementing incrementality using the latest load date per PK (Cost_Center, Company_code, BPP_User, Indicator).

•Created **int_ariba_srm** to standardize and clean the staged data, preserving the required column ordering.

•Developed **int_ariba_approvers** by joining to the company code table and bringing in the **Ariba flag** and related approval attributes.

•Delivered the final data extracts model **gbs_approval** (a.k.a. fp1_approvals) filtered to **ariba = false** for non-Ariba approvals.

•**Outcome achieved:**
A fully validated, incremental, and enriched DBT pipeline was successfully delivered, producing an accurate and properly ordered GBS approval dataset for non-Ariba approvals, with the foundation in place that extends the same framework to Ariba approvals

Key Takeaways

- Established a clean, staging and integration pipeline (stg_ariba_srm, int_ariba_srm, int_ariba_approvers) that preserves required PK logic and column order.

- Delivered a reliable final extract (gbs_approval) that correctly isolates non-Ariba approvals using the Ariba flag.

- Successfully migrated SAP source data into Snowflake using a modern tech stack and automated pipelines.

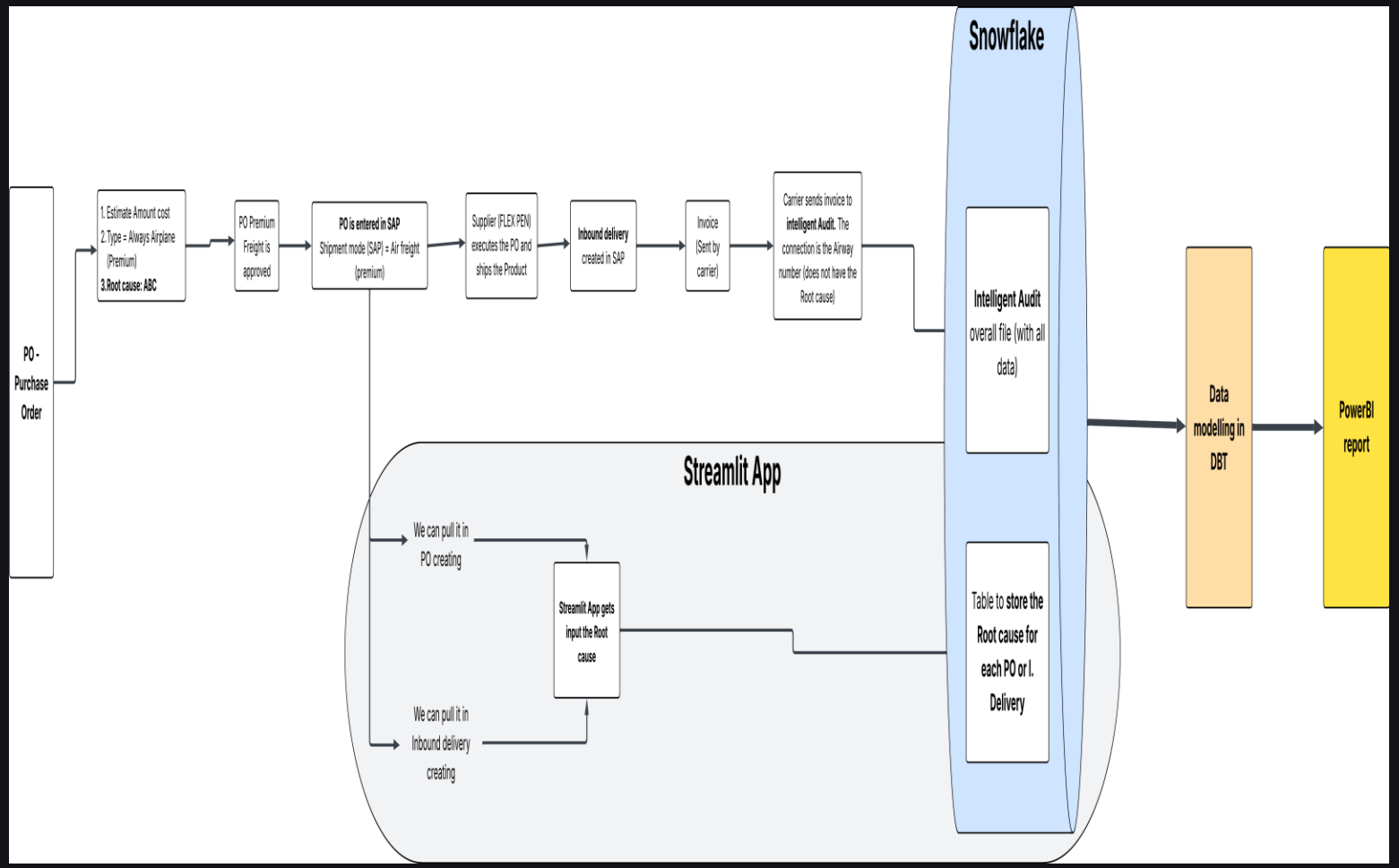
- Ensured data accuracy and usability through a structured DBT workflow that supports future expansion for Ariba approvals.

Premium Freight App

Context of the project

Business Background

- Bose ships goods globally using Land, Sea, or Air. Land is the default, but urgent shipments are sent by Air, incurring "Premium Freight" costs.
- Management has observed a significant increase in Premium Freight spending but lacks visibility into the underlying causes (root causes) for these expedited shipments.
- There is no systematic way to track or analyze why Premium Freight is used, leading to challenges in cost control and process optimization.



App overview

Overview:

- Premium Freight Root Cause Capture system is implemented using a Streamlit application, integrated with Snowflake, SAP data, Intelligent Audit (IA) freight cost files, and ultimately consumed by Power BI for reporting and analytics.

- The purpose of the solution is to transform a historically manual, unstructured, and decentralized process into a standardized, auditable, and analytics-ready workflow that provides full visibility into:

1. Why Premium Freight was booked
2. How much Bose is spending on Premium Freight
3. Which POs, suppliers, customers, plants, products, and processes are the biggest drivers

Functional Scope:

Primary Capabilities
- Capture root cause for any Premium Freight Purchase Order (PO)
- Update root cause and maintain full change history
- Browse, filter, and search categorized and uncategorized POs
- Provide a directory of all captured POs
- Surface root cause data into Snowflake for downstream analytics

Non-Goals
- Changing SAP freight booking rules
- Changing freight type determination logic
- Consolidating Intelligent Audit data
- The application is strictly focused on Root Cause Capture Streamlit application and visibility enablement.

App Architecture

The architecture has four major layers:

A. SAP Layer (Operational System)

SAP is where the Premium Freight process originates.

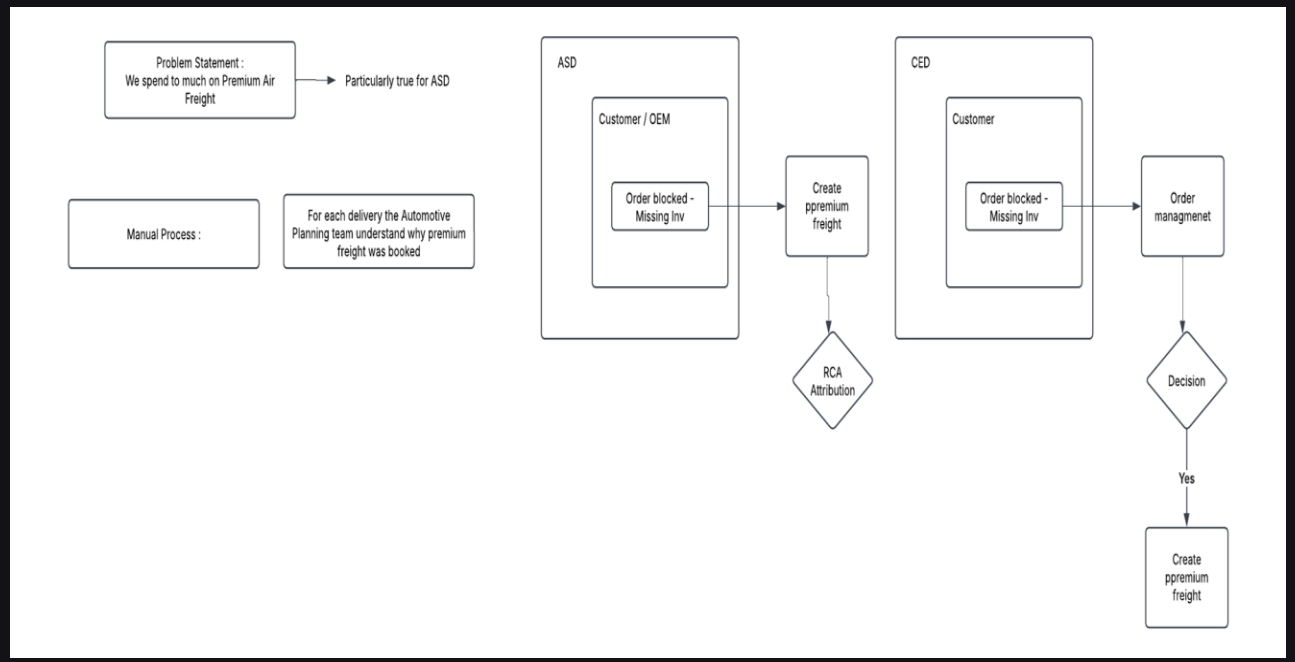
1. PO Creation
Buyer creates a Purchase Order (PO).
Inputs:

- Estimated cost
- Freight type = "Airplane (Premium)"
- **Root Cause** (from Streamlit App)

2. Premium Freight Approval
PO Premium Freight amount is approved.

3. Inbound Delivery

- Supplier ships goods
- Inbound Delivery is created in SAP
- **Streamlit App captures RCA at this step too**



Architecture Layers

B. Streamlit App Layer (User Input Capture)

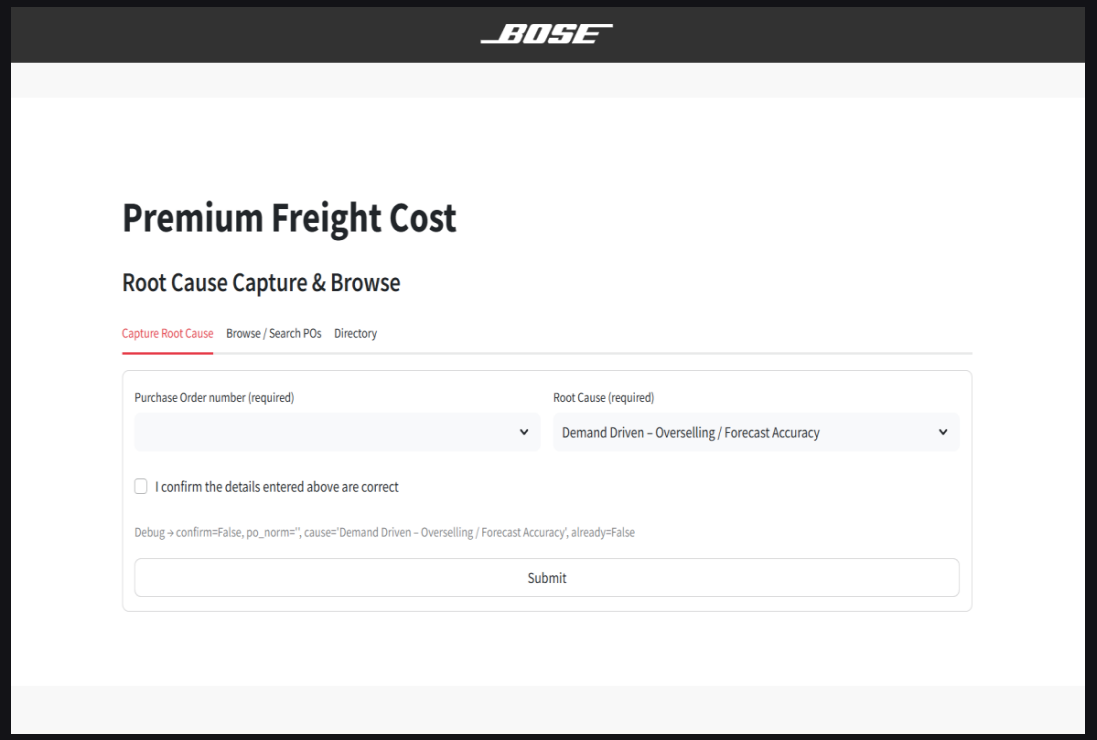
The Streamlit App captures Root Cause Attribution at the right moment:
During PO Creation, it ensures:

- Consistent RCA categories
- Accurate metadata early in the process
- Eliminates manual tracking by Automotive Planning team
- Streamlit → sends structured RCA data to Snowflake table

C. Snowflake Layer (Data Storage & Integration Hub)
Two major datasets:

1. **PO Data – Purchase Order Numbers to check the mode of air freight are fetched from SAP**

2. **Tables using Streamlit app data (From Snowflake) - Root Cause/Reason Codes for the app are fetched from STREAMLIT database inside CONTAINER_FREIGHTPRCOST schema which contains RC_DICT table in Snowflake.**



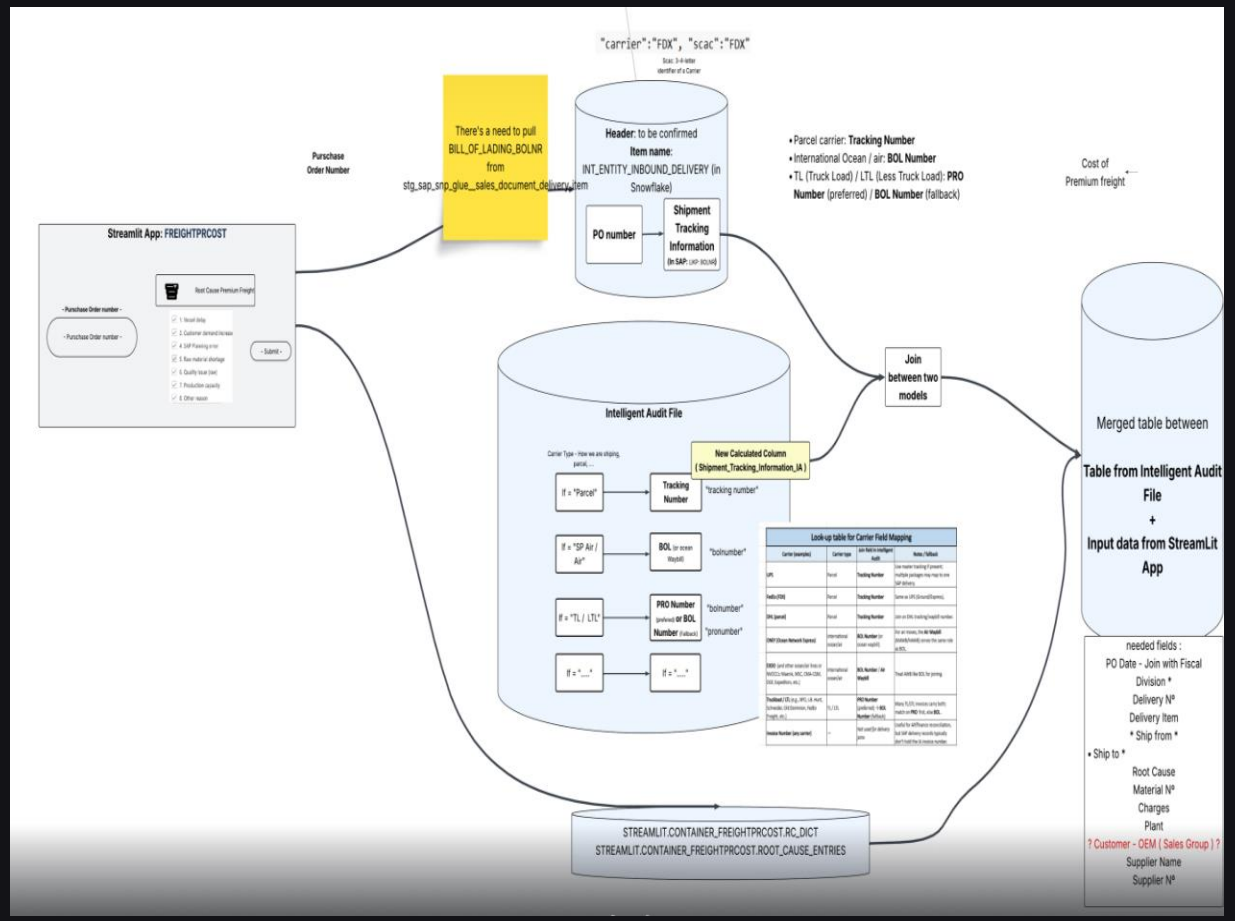
Workflow

- **Premium Freight PO created in SAP**
  SAP identifies freight type (Air) → flagged as Premium Freight.

- **Supplier ships product; carrier invoices via Intelligent Audit**
  Intelligent Audit file includes freight cost, tracking number, BOL/PRO number.

- **SAP + IA data ingested into Snowflake**
- **Planner opens the Streamlit App → Capture**
  - Selects PO from dropdown
  - Selects standardized root cause
  - Confirms and submits

  Streamlit writes records to Snowflake
  - New root cause entry
  - Change history preserved

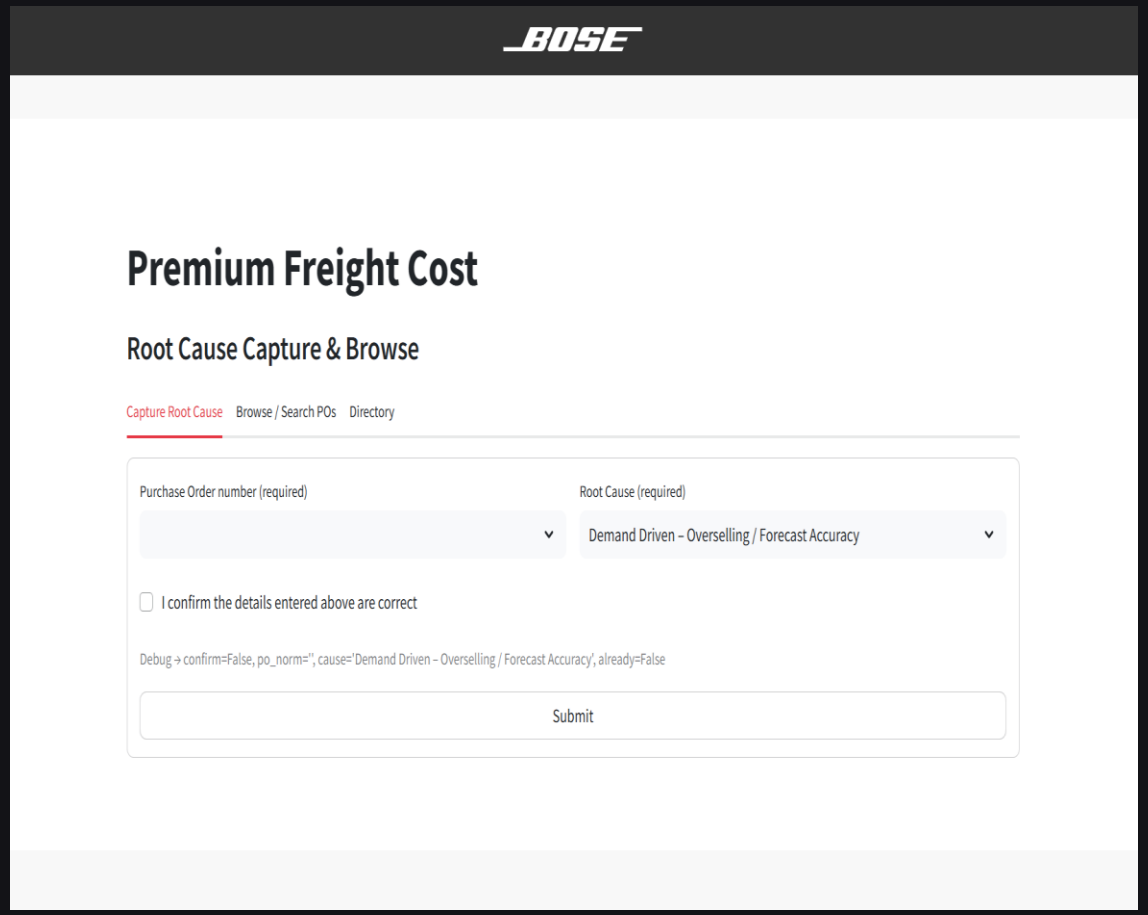


UI Design

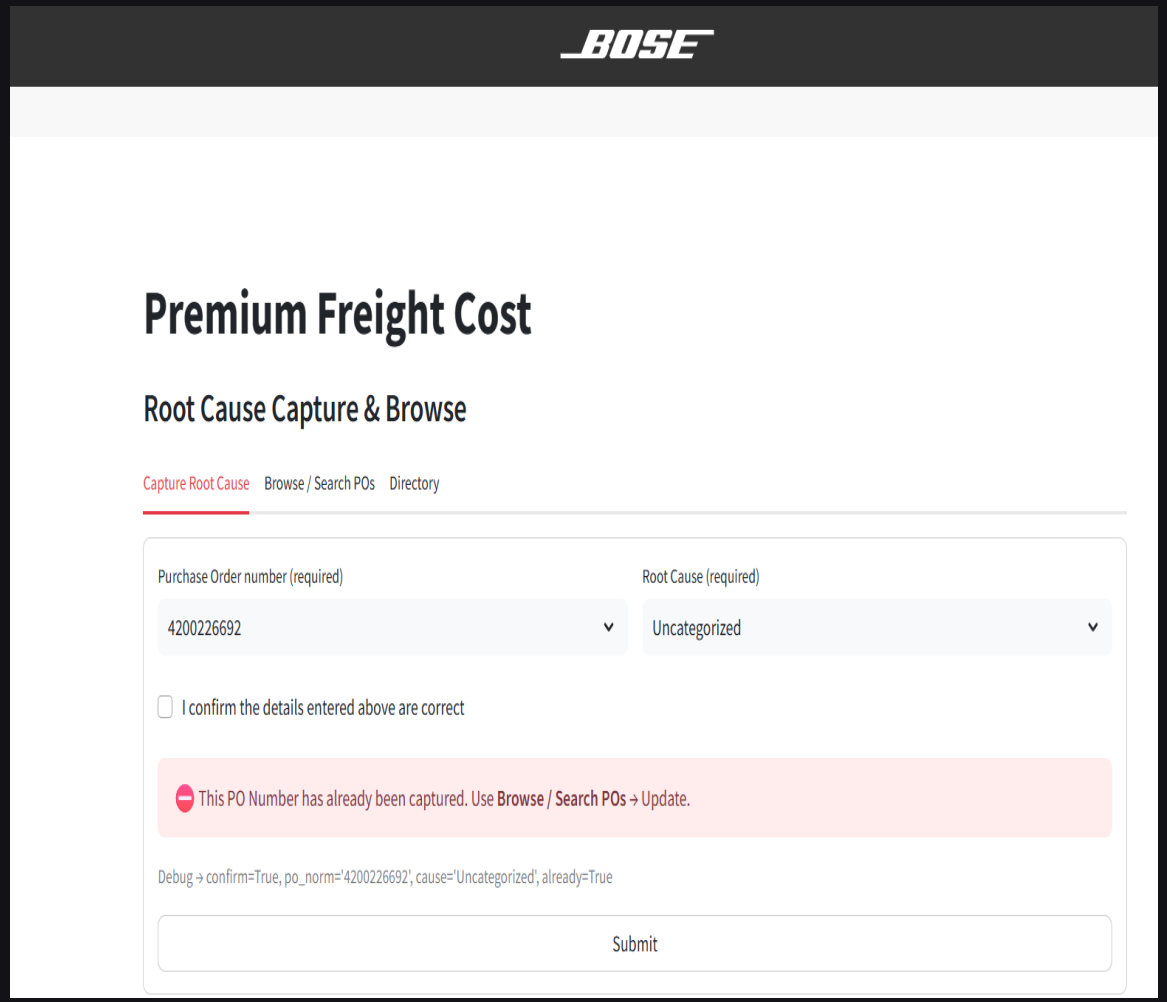
The Streamlit application is structured into **three main tabs**, each designed for a specific part of the workflow.

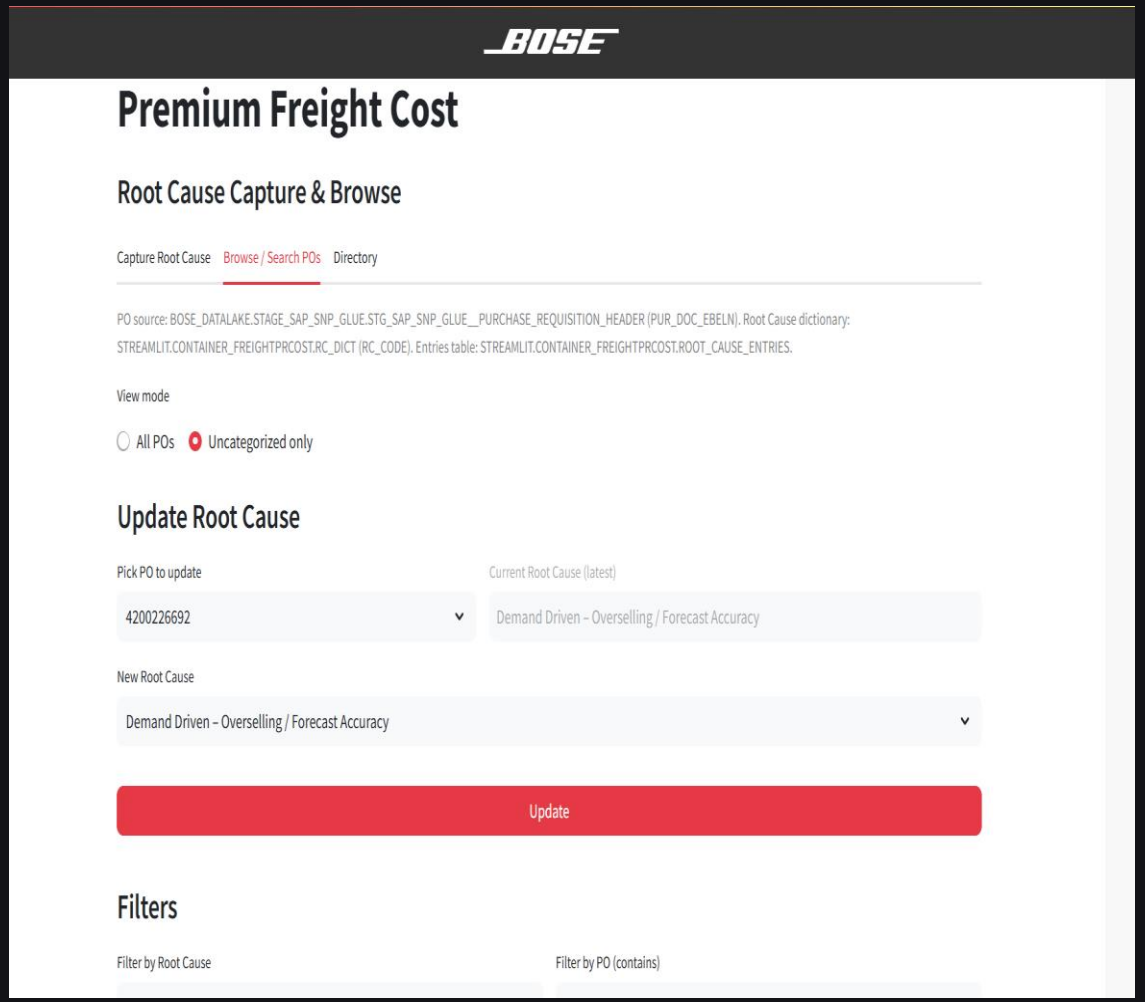
Tab 1 – Capture Root Cause

Element	Purpose
PO Number (Required)	Searchable dropdown filtered to valid Premium Freight POs. Prevents typos and enforces data integrity.
Root Cause (Required)	Dropdown of controlled RCA categories (from Snowflake dictionary).
Confirmation Checkbox	Forces user to verify inputs before submitting.
Submit Button	Writes an entry into the Snowflake ROOT_CAUSE_ENTRIES table.

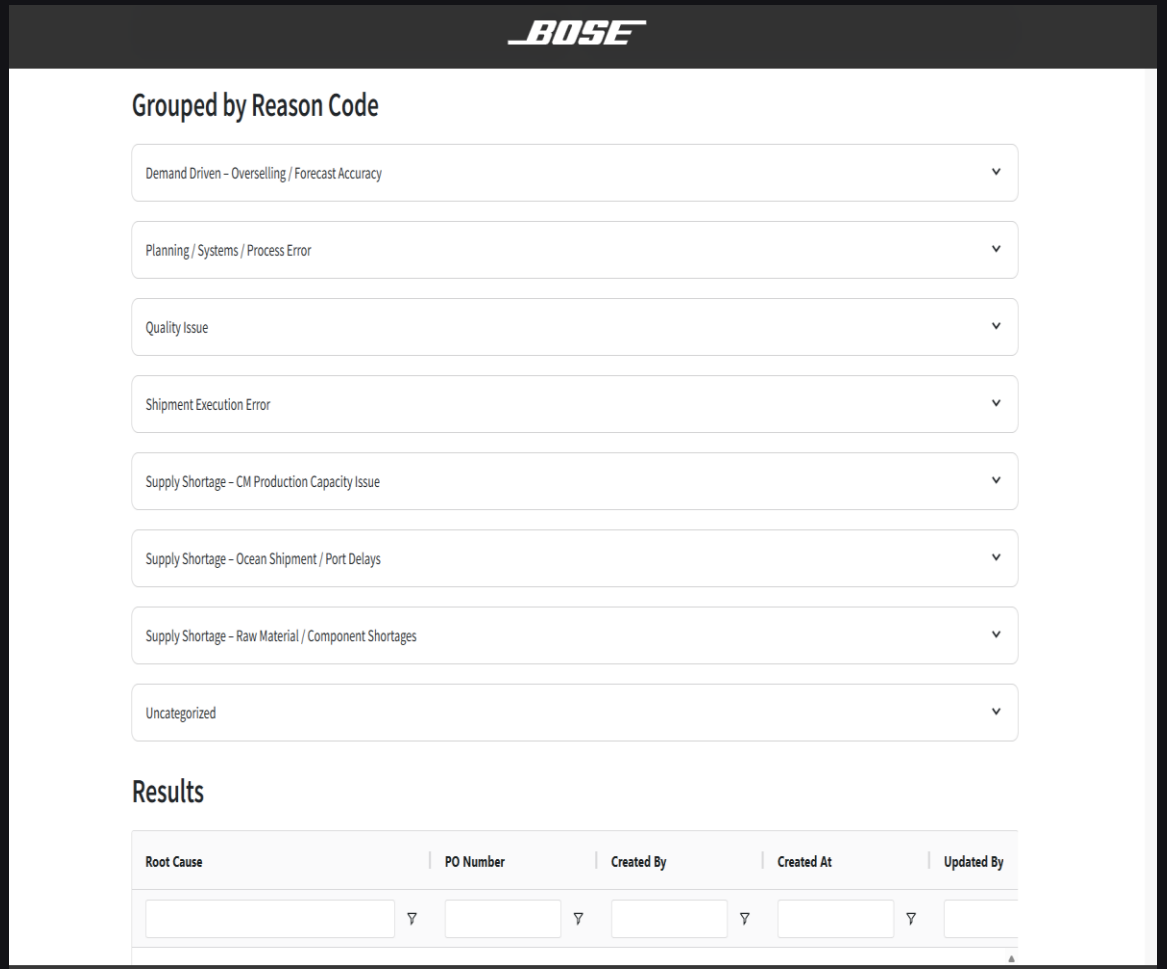


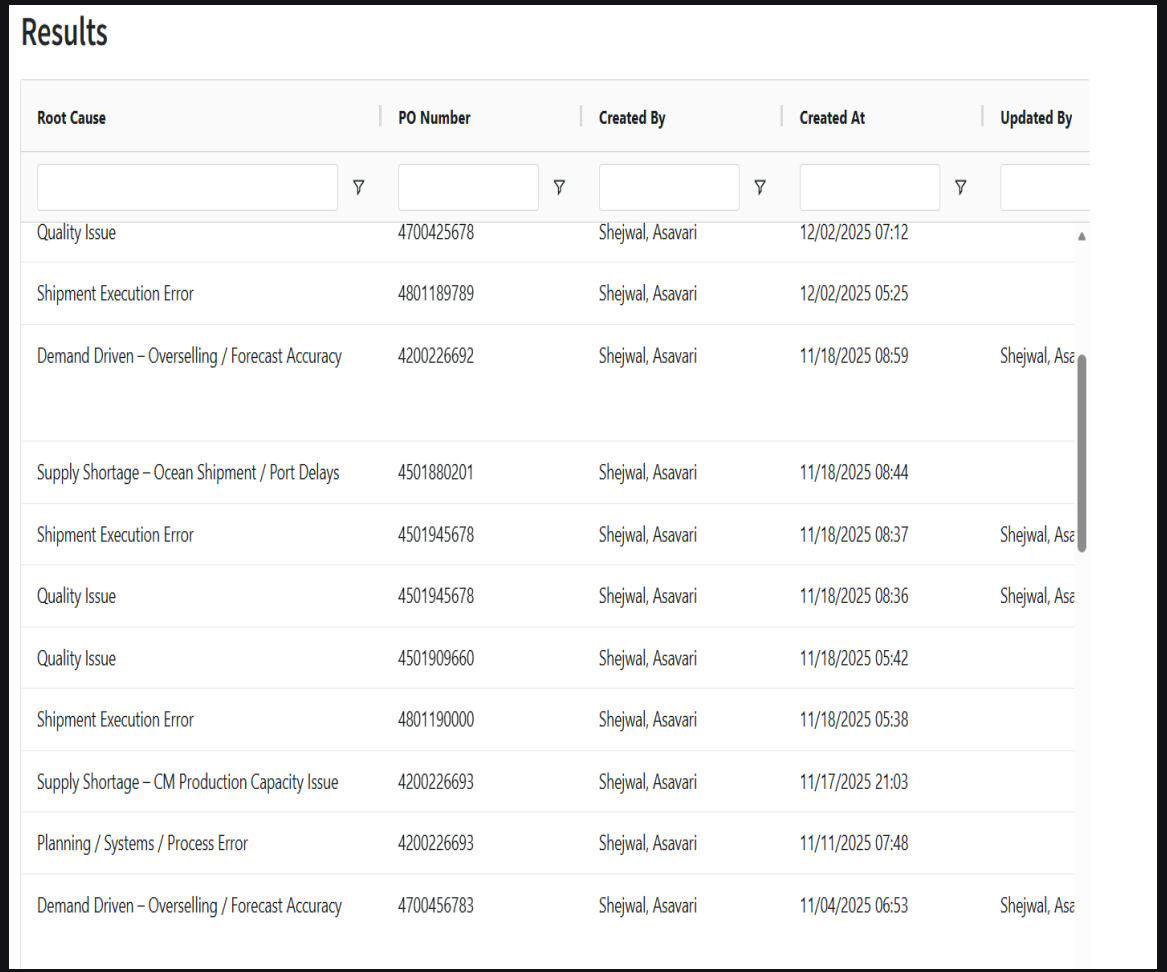
UI Design


BOSE
Premium Freight Cost
Root Cause Capture & Browse
Capture Root Cause
Browse / Search POs
Directory
Purchase Order number (required)
4200226692
Root Cause (required)
Uncategorized
I confirm the details entered above are correct
This PO Number has already been captured. Use Browse / Search POs → Update.
Debug → confirm=True, po_norm='4200226692', cause='Uncategorized', already=True
Submit


BOSE
Premium Freight Cost
Root Cause Capture & Browse
Capture Root Cause
Browse / Search POs
Directory
PO source: BOSE_DATALAKE.STAGE_SAP_SNP_GLUE.STG_SAP_SNP_GLUE__PURCHASE_REQUISITION_HEADER (PUR_DOC_EBELN). Root Cause dictionary: STREAMLIT.CONTAINER_FREIGHTPRCOST.RC_DICT (RC_CODE). Entries table: STREAMLIT.CONTAINER_FREIGHTPRCOST.ROOT_CAUSE_ENTRIES.
View mode
All POs
Uncategorized only
Update Root Cause
Pick PO to update
4200226692
Current Root Cause (latest)
Demand Driven – Overselling / Forecast Accuracy
New Root Cause
Demand Driven – Overselling / Forecast Accuracy
Update
Filters
Filter by Root Cause
Filter by PO (contains)

UI Design





Results

Root Cause	PO Number	Created By	Created At	Updated By
Quality Issue	4700425678	Shejwal, Asavari	12/02/2025 07:12	
Shipment Execution Error	4801189789	Shejwal, Asavari	12/02/2025 05:25	
Demand Driven – Overselling / Forecast Accuracy	4200226692	Shejwal, Asavari	11/18/2025 08:59	Shejwal, Asa
Supply Shortage – Ocean Shipment / Port Delays	4501880201	Shejwal, Asavari	11/18/2025 08:44	
Shipment Execution Error	4501945678	Shejwal, Asavari	11/18/2025 08:37	Shejwal, Asa
Quality Issue	4501945678	Shejwal, Asavari	11/18/2025 08:36	Shejwal, Asa
Quality Issue	4501909660	Shejwal, Asavari	11/18/2025 05:42	
Shipment Execution Error	4801190000	Shejwal, Asavari	11/18/2025 05:38	
Supply Shortage – CM Production Capacity Issue	4200226693	Shejwal, Asavari	11/17/2025 21:03	
Planning / Systems / Process Error	4200226693	Shejwal, Asavari	11/11/2025 07:48	
Demand Driven – Overselling / Forecast Accuracy	4700456783	Shejwal, Asavari	11/04/2025 06:53	Shejwal, Asa

Metrics

Financial Metrics

Total Premium Freight Spend:
Aggregate cost of all shipments classified as Premium Freight.

Cost per Premium Shipment:
Average and distribution of costs per Premium Freight shipment.

Premium Freight Spend as % of Total Freight Spend:
Ratio of Premium Freight costs to total freight costs.

Root Cause Analysis Metrics

Root Cause Distribution: Updated Root Causes. There was an initial list of Root Causes that was used by business

Root Cause nomenclature		
Old	**New**	**Notes**
	Demand Driven	Overselling / Forecast Accuracy (INCLUDES Customer Requirement Increase)
Component / IC Shortage at Luxshare	**Supply Shortage**	Raw Material / Component Shortages (Includes Supplier De-Commits)
Raw Material Shortage	**Supply Shortage**	CM Production Capacity Issue
Vessel Delay	**Supply Shortage**	Ocean Shipment / Port Delays
Other reason	**New Product Introduction**	Planned Air Freight
Quality Issue (Raw)	**Quality Issue**	
SAP Planning error	**Planning / Systems / Process Error**	(Includes “Other” Issues)
N.A.	**Shipment Execution Error**	
N.A.	**Uncategorized**	

Benefits Achieved

Operational Benefits

- Eliminates manual email-based root cause tracking.
- Ensures every Premium Freight PO has a structured explanation.
- Provides full versioning and auditability.

Analytical Benefits

- Enables precise Premium Freight spend attribution.
- Identifies the most common operational or planning issues.
- Supports corrective actions with data-backed evidence.

Strategic Benefits

- Drives accountability and transparency.
- Reduces recurring causes of Premium Freight over time.
- Enables leadership to address systemic issues (forecasting, supplier capacity, execution errors).

Lessons Learned & Next Steps

Lessons Learned

Maintained clean, modular dbt architecture with strong documentation, consistent naming, and reusable logic across staging, intermediate, and mart layers.

Learned to prioritize and discuss data quality by enforcing robust dbt tests, validating source reliability, and using CI/CD to catch issues before deployment.

Learned how to **define** and **measure impact**, ensuring technical work aligns with business goals.

Next Steps

Continue expanding skills in analytical engineering – dbt, Snowflake, data modeling, SQL

Return to Northeastern for one final semester and graduate in April.

Seek opportunities to apply data engineering, and analytical skills to impactful, real-world data challenges that drive business insights and decision-making.

Continue exploring Snowflake Cortex AI, dbt Fusion co-pilot to deepen expertise in AI-driven data engineering capabilities

Thank You !
Q&A